01/ July 2026 (lesson 36: Data Visualization)

Data Visualization

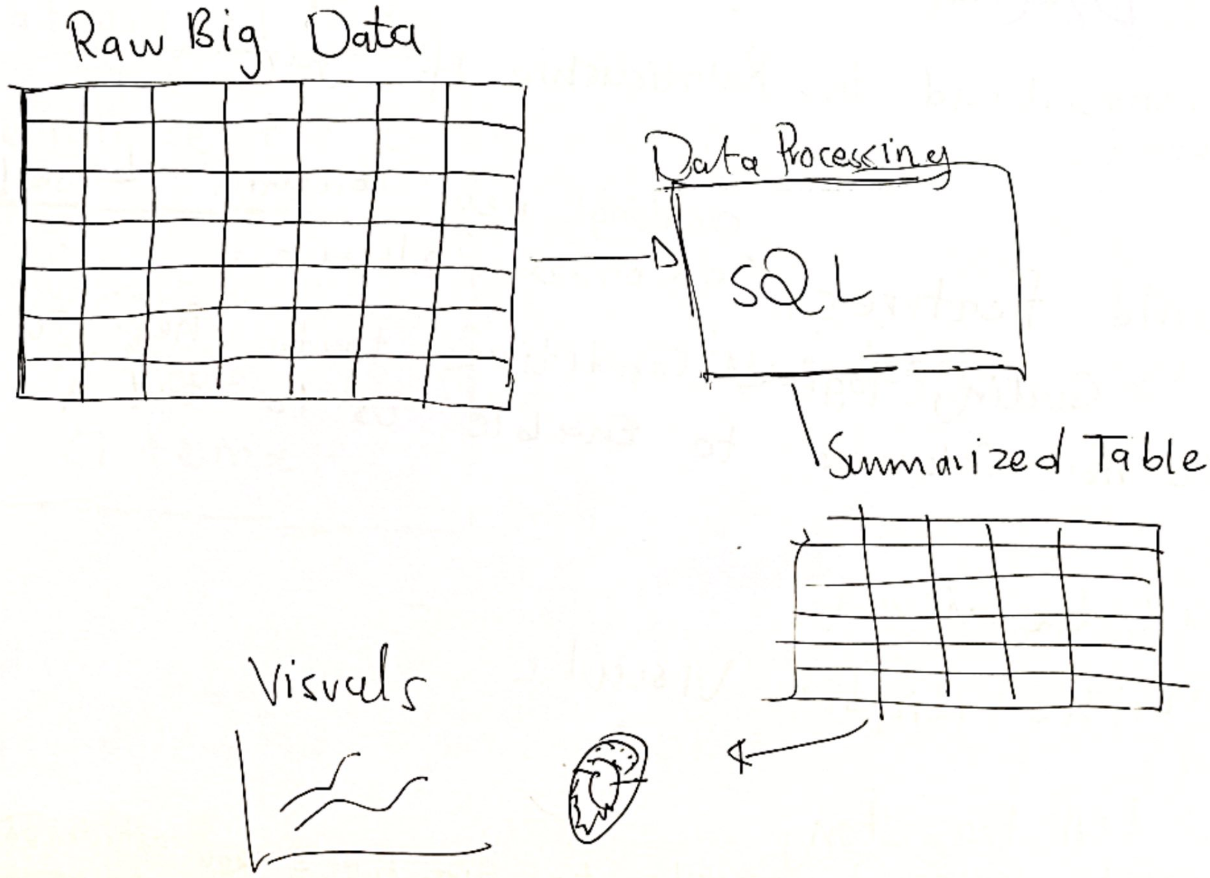
Explaining the Insights from your visuals.

Data story telling

- we build data Visuals to tell a data story
- We build data visuals inorder to Extract Insights from a complex dataset

Insight -> Is Information that I can Extract from the data to use for making business decision.

- We visualize data In order to see the trends



• The way you code, determines Insights you can

Data Analysis Process

1. Understand Business Questions — BRS
   ↳ Business Requirement.
2. Ask data driven questions.
   ↳ you want to set objectives of how you will answer the business goals or business questions.

3. Data Inspection / OLAP Database
   — Do I have all the data that I will use to solve the business problem
   ↳ Is the data I need to solve the business available

4. EDA (Exploratory Data Analysis)
   ↳ understand the Relationship of your data points

5. Build Features — creating new columns based on other columns.
   ↳ Creating logical Conditions that help to create new Columns to Enable us to tell a Story

6. Build the visuals
   ↳ use the Right Visuals.

7. You tell the story
   ↳ Interpreting the visuals to highlight key actionable Insight

Build The Visuals

Categorical Data

Numeric Data

Continous Data

Continuos Data is data that keeps Increasing e.g (age)

quantitative Data

Measuring data or numbers e.g Revenue

Categorical Data

We can use the Categorical data to build filters

Slicer

Gender
- Male
- Female

- splits / legend

Numeric data

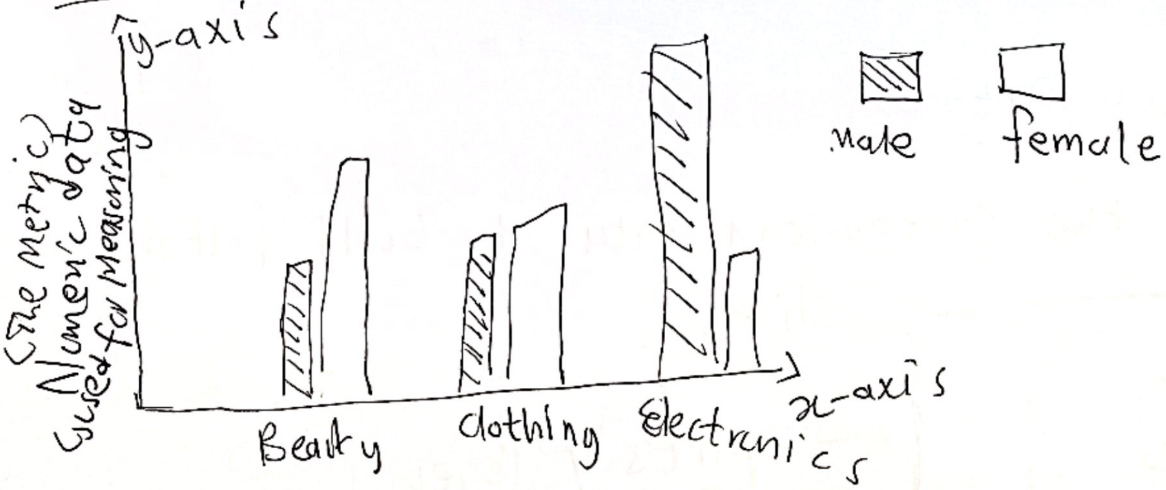
↳ We use numeric data with a card to summarize a specific KPI (key performance Indicator)

Revenue
R100k

Metric to Measure

Categorical Data - helps us group our data
↳ So If you want to filter proper use categorical data to filter or group.

Bar chart



categorical Data
↳ use this to group.

A Measure coud be an ~~opt~~ output of an aggregation.

Data Visualizations

The main settings you see:

- X column → what to group / slice by
- Y column → what to Measure (Ravenue, aggregated sum)
- Group by → further split your data
- Direction

When to use what chart type:

- Pie / Donut → Proportions of a whole
- Bar → Comparing categories
- Line → Trends over time
- Scatter → Relationship between two numeric values.

Simple Rule

X axis (Horizontal) → usually time or sequence
Y axis (Vertical) → usually a numeric value you're measuring

02 July 2026 (lesson 37: Data studio)

Data Studio

Data studio Is a free, web-based data visualization and Reporting tool.

Dimension Refers to like a more categorical data

Metric → Refers to what we are measuring.

A Control Is a filter

Gr. | Activity (Create dashboards)

1. Snowflake
2. Databricks
3. Data Studio.

Build Dashboards on this platforms

Use bright TV dataset.

06 July 2026 (lesson 38: BrightTV Dashboard)

07 July 2026 (Lesson 39: Data Visualization - Data story)

Page 1: Summary Page
↳ Summarize the Important measure

KPIs
cover page
Page 0: Explain Whats In the Dashboard
Page 2: Sectional Detail
↳ Breakdown a specific view Into a detailed view
Page last: Glossary
↳ Definitions of key words
↳ Abbreviations - MoM%
↳ YoY
↳ WOW
↳ MTD
↳ YTD

Note: What Is the lastest data update

Summary Page .

Done (Dashboard Plan)

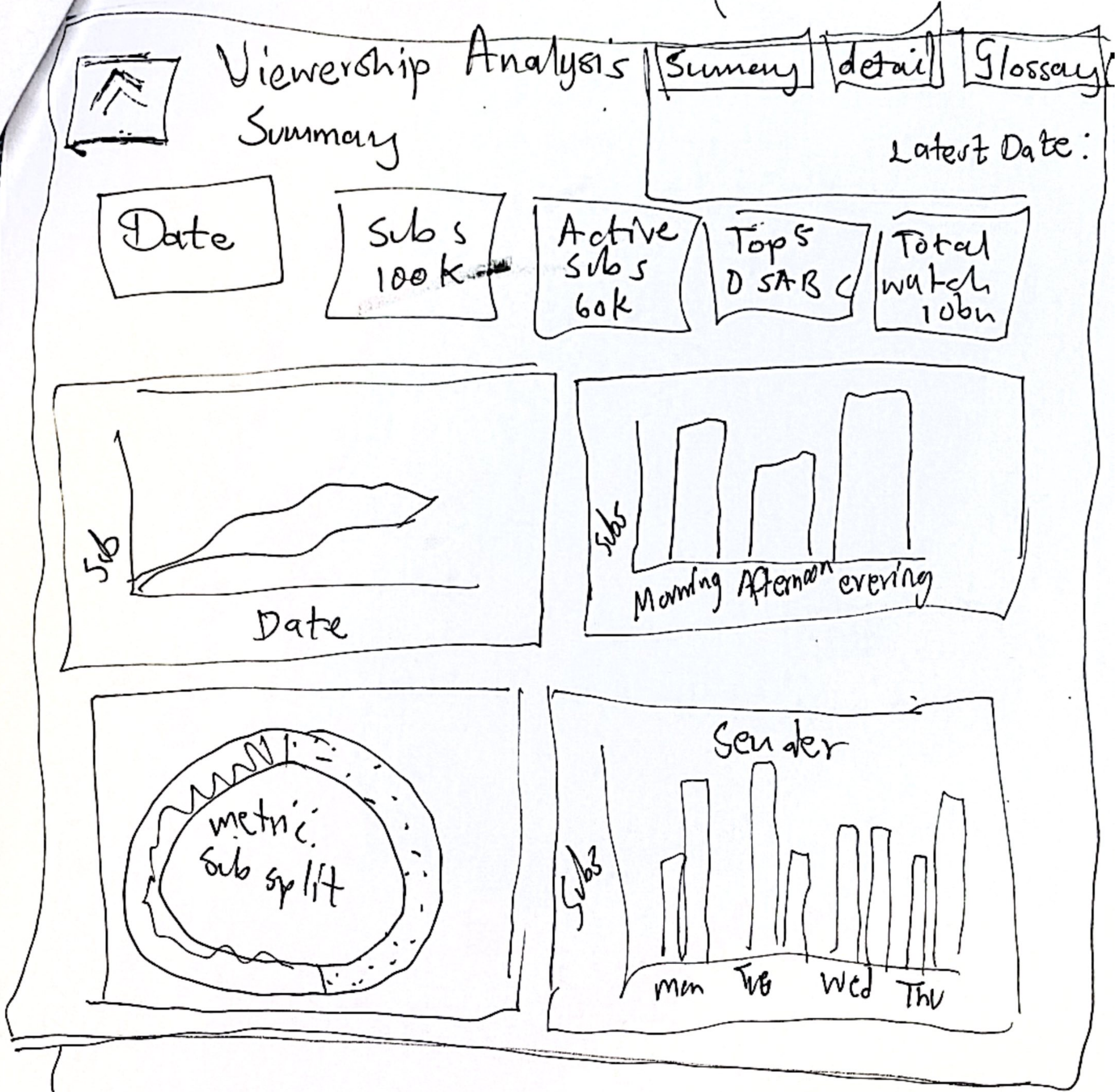
- Build bright Coffee

Lesson 40: No notes (13 July 2027)

Lesson 41 : No notes (14 July 2027)

Summary Page

• Navigation buttons



Page2 : Regional Analysis